

ASPEN® 50 SC

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1. IDENTIFICATION

Product name : ASPEN® 50 SC

Manufacturer or supplier's details

Company : FMC CORPORATION

Address : 2929 WALNUT STREET
PHILADELPHIA, PA 19104 USA
(215) 299-6000 (GENERAL INFORMATION)

E-mail address : SDS-Info@fmc.com

Emergency telephone : +507-8322475
911

Medical Emergency Number : Costa Rica - National Center of Poisoning - (506) 2223-1028;
800-INTOXICA
Dominican Republic: DOMINICAN REPUBLIC - Center for
Drug Information and Poisoning - (809) 562-6601 Ext. 1801
El Salvador - Rosales National Hospital - (503) 2231-9262
Guatemala - Center of Toxicological Information and Assis-
tance - (502) 2251-3560 / 2232-0735
Honduras - Hospital School - (504) 232-6105
Nicaragua - National Center of Toxicology - (505) 2289-4700
ext. 1294 cel. 8755-0983
Panama Center of Research and Information on Medications
and Toxicology (507) 523-4948

Recommended use of the chemical and restrictions on use

Recommended use : Fungicide

Restrictions on use : Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 4

Acute toxicity (Inhalation) : Category 5

Specific target organ toxicity - : Category 2 (Kidney)
repeated exposure (Oral)

Long-term (chronic) aquatic : Category 2
hazard

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GHS label elements

Hazard pictograms



Signal Word

: WARNING

Hazard Statements

: H302 Harmful if swallowed.
H333 May be harmful if inhaled.
H373 May cause damage to organs (Kidney) through prolonged or repeated exposure if swallowed.
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements

Prevention:

P260 Do not breathe mist or vapors.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P273 Avoid release to the environment.

Response:

P301 + P317 + P330 IF SWALLOWED: Get medical help.
Rinse mouth.
P304 + P317 IF INHALED: Get medical help.
P319 Get medical help if you feel unwell.
P391 Collect spillage.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture

: Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
Flutriafol	76674-21-0	$\geq 30 - < 50$
ethanediol	107-21-1	$\geq 2.5 - < 10$
Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates	68412-53-3	$\geq 1 - < 2.5$
1,2-benzisothiazol-3(2H)-one	2634-33-5	$\geq 0.025 - < 0.1$

4. FIRST AID MEASURES

General advice

: Move out of dangerous area.
Show this material safety data sheet to the doctor in attend-

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- ance.
Do not leave the victim unattended.
- If inhaled : Consult a physician after significant exposure.
If unconscious, place in recovery position and seek medical advice.
- In case of skin contact : Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.
- Most important symptoms and effects, both acute and delayed : Harmful if swallowed.
May be harmful if inhaled.
May cause damage to organs through prolonged or repeated exposure if swallowed.
- Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.
- Notes to physician : Treat symptomatically.

5. FIRE-FIGHTING MEASURES

- Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.
- Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Hydrogen fluoride
Nitrogen oxides (NOx)
Carbon oxides
Fluorinated compounds
Hydrogen cyanide
- Specific extinguishing methods : Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

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Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment : Firefighters should wear protective clothing and self-contained for fire-fighters breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for : Never return spills in original containers for re-use.
containment and cleaning up Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against : Normal measures for preventive fire protection.
fire and explosion

Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapors/dust.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Electrical installations / working materials must comply with the technological safety standards.

Further information on stor- : No decomposition if stored and applied as directed.
age stability

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control param- eters / Permissible concentration	Basis

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ethanediol	107-21-1	CPT	50 ppm 100 mg/m3	PA OEL
		Further information: Not classifiable as carcinogenic to humans (appendix 4A)		
		TWA (Vapor)	25 ppm	ACGIH
		STEL (Vapor)	50 ppm	ACGIH
		STEL (Inhalable fraction, Aerosol only)	10 mg/m3	ACGIH

Personal protective equipment

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.

Hand protection
Material : Protective gloves

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Plan first aid action before beginning work with this product.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Color : cream

Odor : slight, alcohol-like

Odor Threshold : No data available

pH : 5.7

Melting point/ range : No data available

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Boiling point/boiling range	:	No data available
Flash point	:	> 105.1 °C
Evaporation rate	:	No data available
Self-ignition	:	ca. 384 °C Method: Tested according to Directive 92/69/EEC.
Upper explosion limit / Upper flammability limit	:	No data available
Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	1.19
Density	:	1.1677 g/cm ³ (23 °C)
Solubility(ies) Water solubility	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available No data available
Viscosity Viscosity, dynamic	:	1,331 mPa.s (20 °C) Method: OECD Test Guideline 114 1,195 mPa.s (40 °C) Method: OECD Test Guideline 114
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive Not explosive

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Oxidizing properties	: Non-oxidizing
Surface tension	: 36 mN/m, 1 g/l, 20 °C, OECD Test Guideline 115 26 mN/m, 25 °C, OECD Test Guideline 115 23 mN/m, 40 °C, OECD Test Guideline 115
Molecular weight	: Not applicable
Particle Size Distribution	: D10 = 0.55 µm ± 0 µm D50 = 1.67 µm ± 0.01 µm D90 = 4.37 µm ± 0.01 µm Measurement technique: laser diffraction

10. STABILITY AND REACTIVITY

Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: No decomposition if stored and applied as directed.
Possibility of hazardous reactions	: No decomposition if stored and applied as directed.
Conditions to avoid	: Avoid extreme temperatures. Avoid formation of aerosol.
Incompatible materials	: Avoid strong acids, bases, and oxidizers.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Harmful if swallowed.
May be harmful if inhaled.

Product:

Acute oral toxicity	: LD50(Rat, female): ca. 550 - 2,000 mg/kg Method: OECD Test Guideline 425 GLP: yes Assessment: The component/mixture is moderately toxic after single ingestion.
Acute inhalation toxicity	: LC50(Rat, male and female): > 5.05 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: Breathing difficulties

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GLP: yes
Remarks: no mortality

Acute dermal toxicity : LD50(Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: no mortality

Components:

Flutriafol:

Acute oral toxicity : LD50 (Rat, female): 300 - 2,000 mg/kg
Method: OECD Test Guideline 423
Target Organs: Liver, Gastrointestinal tract
Symptoms: Fatality
GLP: yes
Assessment: The component/mixture is moderately toxic after single ingestion.

LD50 (Rat, female): 1,030 mg/kg
Method: OECD Test Guideline 425
Target Organs: Liver, Gastrointestinal tract
Symptoms: Breathing difficulties

Acute inhalation toxicity : LC50 (Rat): > 5.2 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Fatality, ataxia, Breathing difficulties
GLP: yes

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Remarks: no mortality

LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 402
Symptoms: Irritation
GLP: yes
Assessment: The substance or mixture has no acute dermal toxicity
Remarks: no mortality

ethanediol:

Acute inhalation toxicity : LC50 (Rat, male and female): > 2.5 mg/l
Exposure time: 6 h
Test atmosphere: dust/mist
Remarks: no mortality

Acute dermal toxicity : LD50 (Mouse, male and female): > 3,500 mg/kg

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Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Acute oral toxicity : LD50 (Rat): > 1,000 mg/kg

Acute dermal toxicity : LD50 (Rabbit): > 2,000 mg/kg

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Not classified based on available information.

Product:

Species : Rabbit
Assessment : No skin irritation
Method : OECD Test Guideline 404
Result : No skin irritation

Components:

Flutriafol:

Species : Rabbit
Assessment : Not classified as irritant

Method : OECD Test Guideline 404
Result : No skin irritation
GLP : yes

Species : Rabbit
Result : No skin irritation

ethanediol:

Species : Rabbit
Result : No skin irritation

Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Result : Skin irritation

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

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Serious eye damage/eye irritation

Not classified based on available information.

Product:

Species	:	Rabbit
Assessment	:	No eye irritation
Method	:	OECD Test Guideline 405
Result	:	No eye irritation
GLP	:	yes

Components:

Flutriafol:

Species	:	Rabbit
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 405
Result	:	slight irritation
GLP	:	yes

Species	:	Rabbit
Assessment	:	Not classified as irritant
Method	:	OECD Test Guideline 405
Result	:	Slight or no eye irritation
GLP	:	yes

Species	:	Rabbit
Result	:	slight irritation

ethanediol:

Species	:	Rabbit
Result	:	No eye irritation

Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Result	:	Irreversible effects on the eye
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1,2-benzisothiazol-3(2H)-one:

Species	:	Bovine cornea
Method	:	OECD Test Guideline 437
Result	:	No eye irritation

Species	:	Rabbit
Method	:	EPA OPP 81-4
Result	:	Irreversible effects on the eye

Respiratory or skin sensitization

Skin sensitization

Not classified based on available information.

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Respiratory sensitization

Not classified based on available information.

Product:

Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Assessment	:	Did not cause sensitization on laboratory animals.
Method	:	OECD Test Guideline 429
Result	:	Did not cause sensitization on laboratory animals.

Components:

Flutriafol:

Test Type	:	Buehler Test
Routes of exposure	:	Skin contact
Species	:	Guinea pig
Assessment	:	Did not cause sensitization on laboratory animals.
Method	:	OECD Test Guideline 406
GLP	:	yes

Species	:	Guinea pig
Result	:	Not a skin sensitizer.

ethanediol:

Test Type	:	Maximization Test
Species	:	Guinea pig
Result	:	Does not cause skin sensitization.

Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Result	:	Substance is not considered to be potential skin sensitiser.
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1,2-benzisothiazol-3(2H)-one:

Test Type	:	Maximization Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.

Species	:	Guinea pig
Method	:	FIFRA 81.06
Result	:	May cause sensitization by skin contact.

Germ cell mutagenicity

Not classified based on available information.

Product:

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative GLP: yes
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Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Components:

Flutriafol:

Genotoxicity in vivo : Test Type: dominant lethal test
Method: OECD Test Guideline 478
Result: negative

ethanediol:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OPPTS 870.5100
Result: negative

Genotoxicity in vivo : Test Type: dominant lethal test
Species: Rat
Application Route: Oral
Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro : Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: positive

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Rat (male)
Cell type: Liver cells
Application Route: Ingestion
Exposure time: 4 h
Method: OECD Test Guideline 486
Result: negative

Test Type: Micronucleus test
Species: Mouse
Application Route: Oral
Method: OECD Test Guideline 474

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Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Not classified based on available information.

Components:

Flutriafol:

Species : Mouse
Exposure time : 2 Years
NOAEL : 1.2 mg/kg bw/day
Result : negative

Species : Rat
Exposure time : 2 Years
NOAEL : 1 mg/kg bw/day
Result : negative

Carcinogenicity - Assessment : Animal testing did not show any carcinogenic effects.

ethanediol:

Species : Mouse
Application Route : Oral
Exposure time : 24 month(s)
Result : negative

Reproductive toxicity

Not classified based on available information.

Components:

Flutriafol:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Method: OECD Test Guideline 416
Result: negative

Effects on fetal development : Test Type: Embryo-fetal development
Method: OECD Test Guideline 414
Result: negative

1,2-benzisothiazol-3(2H)-one:

Effects on fertility : Species: Rat, male
Application Route: Ingestion
General Toxicity Parent: NOAEL: 18.5 mg/kg body weight
General Toxicity F1: NOAEL: 48 mg/kg body weight
Fertility: NOAEL: 112 mg/kg bw/day
Symptoms: No effects on reproduction parameters.

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Method: OPPTS 870.3800

Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

Not classified based on available information.

Components:

Flutriafol:

Assessment : May cause respiratory irritation.

STOT-repeated exposure

May cause damage to organs (Kidney) through prolonged or repeated exposure if swallowed.

Components:

ethanediol:

Routes of exposure : Oral
Target Organs : Kidney
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

1,2-benzisothiazol-3(2H)-one:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Repeated dose toxicity

Components:

Flutriafol:

Species : Rat
NOAEL : 13.3 mg/kg bw/day
Application Route : Oral - feed
Exposure time : 90 d
Symptoms : anemia, Liver effects

Species : Dog
NOAEL : 5 mg/kg bw/day
Application Route : Oral
Exposure time : 90 d
Symptoms : blood effects, Liver effects

ethanediol:

Species : Rat
NOAEL : 150 mg/kg
Application Route : Oral

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Exposure time : 12 Months

Species : Dog

NOAEL : > 2,200 - < 4,400 mg/kg

Application Route : Dermal

Exposure time : 4 Weeks

Method : OECD Test Guideline 410

1,2-benzisothiazol-3(2H)-one:

Species : Rat, male and female

NOAEL : 15 mg/kg

Application Route : Ingestion

Exposure time : 28 d

Method : OECD Test Guideline 407

Symptoms : Irritation

Species : Rat, male and female

NOAEL : 69 mg/kg

Application Route : Ingestion

Exposure time : 90 d

Symptoms : Irritation, Reduced body weight

Aspiration toxicity

Not classified based on available information.

Components:

Flutriafol:

The substance does not have properties associated with aspiration hazard potential.

Neurological effects

Components:

Flutriafol:

No neurotoxicity observed in animal studies.

Further information

Product:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 335.9 mg/l

Exposure time: 96 h

Test Type: static test

Method: OECD Test Guideline 203

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Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 112.1 mg/l
End point: Immobilization
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (algae)): 208.8 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

EyC50 (Pseudokirchneriella subcapitata (algae)): 14.5 mg/l
Method: OECD Test Guideline 201

Toxicity to soil dwelling organisms : > 1,000 mg/kg
Exposure time: 14 d
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP: yes

Toxicity to terrestrial organisms : LD50: 720 mg/kg
Species: Coturnix japonica (Japanese quail)
Method: US EPA Test Guideline OPPTS 850.2100
GLP: yes

LD50: > 200 µg/bee
Species: Apis mellifera (bees)
Method: OECD Test Guideline 214
Remarks: Contact

LD50: 247 µg/bee
Species: Apis mellifera (bees)
Method: OECD Test Guideline 213
Remarks: Oral

Components:

Flutriafol:

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 33 mg/l
Exposure time: 96 h

LC50 (Danio rerio (zebra fish)): 22.97 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 67 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes

EC50 (Daphnia magna (Water flea)): 42.21 mg/l
End point: Immobilization

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	Exposure time: 48 h Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	: IC50 (<i>Selenastrum capricornutum</i> (green algae)): 12 mg/l Exposure time: 96 h IC50 (<i>Scenedesmus subspicatus</i>): 1.9 mg/l Exposure time: 72 h EbC50 (<i>Lemna gibba</i> (duckweed)): 0.65 mg/l Exposure time: 7 d EyC50 (<i>Pseudokirchneriella subcapitata</i> (algae)): 3.69 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 GLP: yes
Toxicity to fish (Chronic toxicity)	: NOEC: 4.8 mg/l Exposure time: 28 d Species: <i>Lepomis macrochirus</i> (Bluegill sunfish) NOEC: 20 mg/l Exposure time: 14 d Species: <i>Danio rerio</i> (zebra fish) Method: OECD Test Guideline 204
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC: 0.31 mg/l Exposure time: 21 d Species: <i>Daphnia magna</i> (Water flea) NOEC: 0.45 mg/l Exposure time: 21 d Species: <i>Daphnia magna</i> (Water flea) Method: OECD Test Guideline 211
Toxicity to soil dwelling organisms	: NOEC: 0.01 mg/cm ² Exposure time: 180 d Species: <i>Eisenia fetida</i> (earthworms) LC50: > 1,000 mg/kg Exposure time: 14 d Species: <i>Eisenia fetida</i> (earthworms) Method: OECD Test Guideline 207
Toxicity to terrestrial organisms	: LD50: > 144 µg/bee End point: Acute oral toxicity Species: <i>Apis mellifera</i> (bees) Method: OECD Test Guideline 213 GLP: yes LD50: > 150 µg/bee End point: Acute contact toxicity Species: <i>Apis mellifera</i> (bees)

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Method: OECD Test Guideline 214
GLP: yes

LD50: > 100 µg/bee
End point: Acute contact toxicity
Species: Apis mellifera (bees)
Method: OECD Test Guideline 214

LD50: 872.53 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity
Species: Apis mellifera (bees)
Method: OECD Test Guideline 213

LD50: > 5,000 mg/kg
Species: Anas platyrhynchos (Mallard duck)

LD50: ca. 385 mg/kg
Species: Coturnix japonica (Japanese quail)
Method: US EPA Test Guideline OPPTS 850.2100

LD50: 4260 ppm
Species: Coturnix japonica (Japanese quail)
Method: OPPTS 850.2200

ethanediol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): > 72,860 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : IC50 (Pseudokirchneriella subcapitata (green algae)): 10,940 mg/l
Exposure time: 96 h

Toxicity to microorganisms : (activated sludge): > 1,995 mg/l
Exposure time: 30 min
Method: ISO 8192

Toxicity to fish (Chronic toxicity) : 1,500 mg/l
Exposure time: 28 d
Species: Menidia peninsulæ (tidewater silverside)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : 33,911 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

Poly(oxy-1,2-ethanediyl), α-(nonylphenyl)-ω-hydroxy-, branched, phosphates:

Toxicity to fish : LC50 (Fish): > 1 mg/l
Exposure time: 96 h

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Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 1 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : EC50 (algae): > 1 mg/l
Exposure time: 72 h

Toxicity to microorganisms : Remarks: No data available

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish : LC50 (Cyprinodon variegatus (sheepshead minnow)): 16.7 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 2.15 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 2.9 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 0.070 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Pseudokirchneriella subcapitata (green algae)): 0.04 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

M-Factor (Acute aquatic toxicity) : 10

Toxicity to microorganisms : EC50 (activated sludge): 24 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

EC50 (activated sludge): 12.8 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

Persistence and degradability

Components:

Flutriafol:

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Biodegradability : Result: Not readily biodegradable.

Stability in water : Remarks: Does not readily hydrolyze

ethanediol:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 90 - 100 %
Exposure time: 10 d
Method: OECD Test Guideline 301A

Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Biodegradability : Result: Inherently biodegradable.

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable
Method: OECD Test Guideline 301C

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Remarks: No data available

Components:

Flutriafol:

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 7
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 2.29

ethanediol:

Partition coefficient: n-octanol/water : log Pow: -1.36

Poly(oxy-1,2-ethanediyl), α -(nonylphenyl)- ω -hydroxy-, branched, phosphates:

Bioaccumulation : Remarks: No data available

Partition coefficient: n-octanol/water : Remarks: No data available

1,2-benzisothiazol-3(2H)-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Exposure time: 56 d

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Bioconcentration factor (BCF): 6.62
Method: OECD Test Guideline 305
Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).

Partition coefficient: n-octanol/water : log Pow: 0.7 (20 °C)
pH: 7

log Pow: 0.99 (20 °C)
pH: 5

Mobility in soil

Components:

Flutriafol:

Distribution among environmental compartments : Remarks: Moderately mobile in soils

Stability in soil : Remarks: Very persistent in soil.

1,2-benzisothiazol-3(2H)-one:

Distribution among environmental compartments : Koc: 9.33 ml/g, log Koc: 0.97
Method: OECD Test Guideline 121
Remarks: Highly mobile in soils

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

Components:

Flutriafol:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

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Contaminated packaging : It is prohibited to reuse, bury, burn, or sell containers. Rinsable containers: Triple rinse containers of less than 20 liters and pressure rinse containers of 20 liters or more. Triple rinsing: Add water up to $\frac{1}{4}$ of the container's capacity, close and shake for 30 seconds. Pour the rinse water into the mixing tank, considering this volume of water within the recommended volume for mixing preparation. Perform this procedure three times. Pressure rinsing: Activate the pressure rinsing device for 30 seconds, considering the volume of water used as part of the recommended volume for mixing preparation. In both procedures, punctured the container on its base without damaging the label. In all cases, take the empty containers to collection points indicated by the local empty containers program.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Flutriafol)

Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s. (Flutriafol)

Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Flutriafol)

Class : 9
Packing group : III
Labels : 9

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EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to IMO instruments

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

List of Precursors and Controlled Chemicals. : sodium hydroxide

The ingredients of this product are reported in the following inventories:

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. Flutriafol
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

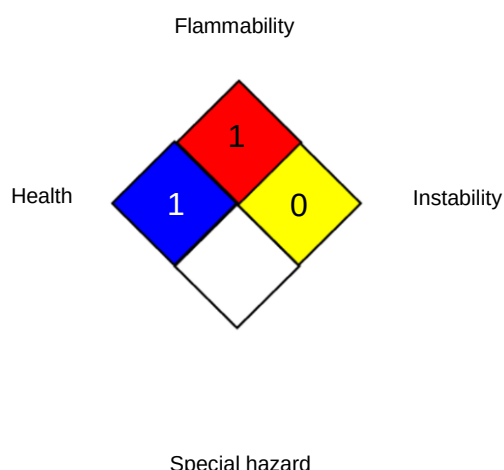
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Date format : dd.mm.yyyy

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Further information

NFPA:



HMIS® IV:

HEALTH	*	2
FLAMMABILITY		1
PHYSICAL HAZARD		0

HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

Full text of other abbreviations

ACGIH	:	USA. ACGIH Threshold Limit Values (TLV)
PA OEL	:	Panama. Occupational Exposure Limits
ACGIH / TWA	:	8-hour, time-weighted average
ACGIH / STEL	:	Short-term exposure limit
PA OEL / CPT	:	Time Weighted Concentration (8 hours of exposure)

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances

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es; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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